

Binomial Distribution

$$P(y|\theta) = \binom{n}{y} \theta^y (1-\theta)^{n-y}$$

$$\binom{n}{y} = \frac{n!}{y!(n-y)!}$$



$$\theta = 0.75 \quad \begin{matrix} H=1 \\ T=0 \end{matrix}$$

What is the probability of TTT?

$$P(y=0, n=3 | \theta=0.75) = \binom{3}{0} (0.75)^0 (0.25)^3 \approx 1.6\%$$

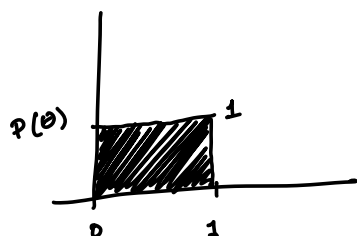
$$\rightarrow P(y=1, n=3 | \theta=0.75) = \binom{3}{1} (0.75)^1 (0.25)^2 \approx 14\%$$

$$\rightarrow \{ TTH, THT, HTT \}$$

Binomial Likelihood:

$$P(y|\theta) = \binom{n}{y} \theta^y (1-\theta)^{n-y}$$

Prior:



$$P(\theta) = 1 \quad 0 \leq \theta \leq 1$$

Posterior:

$$P(\theta|y) \propto P(\theta) P(y|\theta)$$

constant w.r.t. θ

$$= \binom{n}{y} \theta^y (1-\theta)^{n-y}$$

$$0 \leq \theta \leq 1$$

$$P(\theta|y) \propto \underline{\theta^y (1-\theta)^{n-y}}$$

once normalized

$$P(\theta|y) = \frac{B(y+1, n-y+1)}{\int \frac{B(y+1, n-y+1)}{p}}$$

$$p(\tilde{y}|y) = \int p(\tilde{y}, \theta|y) d\theta$$

$$= \int p(\tilde{y}|\theta, y) p(\theta|y) d\theta$$

chain rule of joint probabilities

$$= \int \underbrace{p(\tilde{y}|\theta)}_{\text{prediction}} \underbrace{p(\theta|y)}_{\text{posterior}} d\theta$$

→ posterior predictive distribution

$$p(\tilde{y}|\theta) = \binom{m}{\tilde{y}} \theta^{\tilde{y}} (1-\theta)^{m-\tilde{y}}$$

$$p(\theta|y) = \text{Beta}(y+1, n-y+1) = \frac{\Gamma(n+2)}{\Gamma(y+1)\Gamma(n-y+1)} \theta^y (1-\theta)^{n-y}$$

$$p(\tilde{y}|y) = \int \frac{\Gamma(n+2)}{\Gamma(y+1)\Gamma(n-y+1)} \cdot \binom{m}{\tilde{y}} \cdot \theta^{y+\tilde{y}} (1-\theta)^{m+n-(\tilde{y}+y)} d\theta$$

$$= \binom{m}{\tilde{y}} \frac{\Gamma(n+2)}{\Gamma(y+1)\Gamma(n-y+1)} \cdot \frac{\Gamma(y+\tilde{y}+1) \Gamma(m+n-\tilde{y}-y+1)}{\Gamma(m+n+2)} \rightarrow 1$$

$$\times \int \frac{\Gamma(m+n+2)}{\Gamma(y+\tilde{y}+1) \Gamma(m+n-\tilde{y}-y+1)} \theta^{y+\tilde{y}} (1-\theta)^{m+n-(\tilde{y}+y)} d\theta$$

Beta distribution

$$= \binom{m}{\tilde{y}} \frac{\Gamma(n+2)}{\Gamma(y+1)\Gamma(n-y+1)} \frac{\Gamma(y+\tilde{y}+1) \Gamma(m+n-\tilde{y}-y+1)}{\Gamma(m+n+2)} \cdot 1$$

$$= \binom{m}{\tilde{y}} \frac{(n+1)! (y+\tilde{y})! (m+n-\tilde{y}-y)!}{y! (n-y)! (m+n+1)!}$$

$$\begin{aligned}
&= \binom{m}{\tilde{y}} \frac{(n+1)! (y+\tilde{y})! (m+n-\tilde{y}-y)!}{y! (n-y)! (m+n+1)!} \\
&= \binom{m}{\tilde{y}} \frac{(n+1)}{m+n+1} \cdot \frac{n!}{y!(n-y)!} \cdot \frac{(y+\tilde{y})! (m+n-(y+\tilde{y}))!}{(m+n)!} \\
&= \binom{m}{\tilde{y}} \binom{n}{y} \left[\frac{m+n}{y+\tilde{y}} \right]^{-1} \cdot \frac{n+1}{m+n+1}
\end{aligned}$$

n = # of data points
 y = # of successes

m = # of predicted points
 $\tilde{y}$ = # of predicted successes

Prediction: next birth is female $m = \tilde{y} = 1$

$$\begin{aligned}
p(\tilde{y}=1, m=1 | y) &= \binom{n}{y} \left[\frac{n+1}{y+1} \right]^{-1} \cdot \frac{n+1}{n+2} \\
&= \frac{n!}{y!(n-y)!} \cdot \frac{y+1}{(n+1)!} \cdot \frac{n+1}{n+2} = \frac{y+1}{n+2} \\
&= \frac{y+1}{n+2}
\end{aligned}$$

data
 ↓
 mean of posterior

$$\frac{y+1}{n+2}$$

Beta(α, β) have a mean of $\frac{\alpha}{\alpha + \beta}$

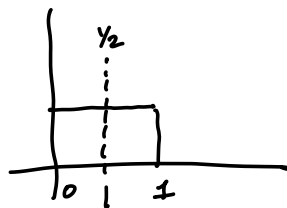
What is the mean of the posterior distribution?

$$p(\theta|y) = \text{Beta}(y+1, n-y+1)$$

$E(\theta|y)$ is the mean

$$= \frac{y+1}{n-y+1+y+1} = \frac{y+1}{n+2}$$

if n is close to 0, then the average is more influenced by the average of the posterior distribution



As n gets large,

$$E(\theta|y) \sim \frac{y}{n}$$